

A02

Q1: Statistics Proof

$$\begin{aligned} 1) \quad m(a+bX) &= a+b(m(X)) \\ m(a) + m(bX) &= a+b(m(X)) \\ a + b(m(X)) &= a+b(m(X)) \quad \checkmark \end{aligned}$$

$$\begin{aligned} 2) \quad \text{Cov}(X, a+bY) &= b \times \text{Cov}(X, Y) \\ \text{Cov}(X, a) + \text{Cov}(X, bY) &= b \times \text{Cov}(X, Y) \\ 0 + \text{Cov}(X, bY) &= b \times \text{Cov}(X, Y) \\ b \times \text{Cov}(X, Y) &= b \times \text{Cov}(X, Y) \quad \checkmark \end{aligned}$$

$$\begin{aligned} 3) \quad \text{Cov}(a+bX, a+bX) &= b^2 \text{Cov}(X, X) \\ \text{Cov}(a+bX, a) + \text{Cov}(a+bX, bX) &= b^2 \text{Cov}(X, X) \\ \text{Cov}(a, a) + \text{Cov}(a, bX) + \text{Cov}(a, bX) + \text{Cov}(bX, bX) &= b^2 \text{Cov}(X, X) \\ 0 + 0 + 0 + \text{Cov}(bX, bX) &= b^2 \text{Cov}(X, X) \\ b \cdot b \text{Cov}(X, X) &= b^2 \text{Cov}(X, X) \\ b^2 \text{Cov}(X, X) &= b^2 \text{Cov}(X, X) \quad \checkmark \end{aligned}$$

Pr 2: $\text{Cov}(X, X) = s^2$

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(X)) (x_i - m(X)) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2$$

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4) $m(g(X)) = g(m(X))$
No it is not always.